Solutions to Exercises from Introduction To Set Theory

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1 Sets

1.3 The Axioms

Exercise 1.3.7. Prove the Axiom of Pair, the Axiom of Union, and the Axiom of Power Set using their weaker versions.

- Weak Axiom of Pair. For any A and B, there is a set C such that $A \in C$ and $B \in C$.
- Weak Axiom of Union. For any S, there exists U such that if $x \in A$ and $A \in S$, then $x \in U$.
- Weak Axiom of Power Set. For any set S, there exists a P such that $x \subseteq S$ implies $x \in P$.

Proof.

- The Axiom of Pair. Let A and B be sets, and let C be a set such that $A \in C$ and $B \in C$. Now consider the set $Y = \{x \in C : x = A \text{ or } x = B\}$. We show that $x \in Y$ if and only if $x = A$ or $x = B$. Indeed, $x = A$ or $x = B$ for any $x \in Y$. Conversely, if $x = A$ or $x = B$, then $x \in C$ by the definition of C, giving $x \in Y$. Hence Y has the required property.
- The Axiom of Union. Let S be a set. By the weak axiom of union, there exists a U such that for every x and A, $x \in U$ if $x \in A$ and $A \in S$. Define $Y = \{x \in U : x \in A \text{ and } A \in S \text{ for some } A\}$. Clearly, if $x \in Y$, then $x \in A$ and $A \in S$ for some A. Conversely, if x is a set such that $x \in A$ and $A \in S$ for some A, then $x \in U$, by definition. Hence $x \in Y$. Therefore, for any S, there is a Y such that $x \in Y$ if and only if $x \in A$ and $x \in S$ for some A.
- The Axiom of Power Set. Let S be a set, and let P be the set implied by the weak axiom of power set. Consider the set $Y = \{x \in P : x \subseteq S\}$. Since $x \subseteq S$ implies $x \in P$, the statements "$x \subseteq S$" and "$x \in P$ and $x \subseteq S$" are equivalent. Hence we have $x \in Y \leftrightarrow x \in P$ and $x \subseteq S \leftrightarrow x \subseteq S$. Therefore Y satisfies the required property.

□

1.4 Elementary Operations on Sets

Exercise 1.4.4. Let A be a set; show that a "complement" of A does not exist. (The "complement" of A is the set of all $x \notin A$.)

Proof. Let A be a set. Suppose that there exists a set X such that $x \in X$ if and only if $x \notin A$. Now consider the set $A \cup X$. For every x, $x \in A \cup X$

if and only if $x \in A$ or $x \in X$, i.e., if and only if $x \in A$ or $x \notin A$. That implies $x \in A \cup X$ for every x and thus $A \cup X$ is the universal set. But this contradicts Exercise 3.3, which states no universal set exists. Therefore the "complement" of A does not exist. □

Exercise 1.4.5. Let $S \neq \varnothing$ and A be sets.

a) Set $T_1 = \{Y \in \mathcal{P}(A) : Y = A \cap X \text{ for some } X \in S\}$, and prove $A \cap \bigcup S = \bigcup T_1$ (generalized distributed law).

b) Set $T_2 = \{Y \in \mathcal{P}(A) : Y = A - X \text{ for some } X \in S\}$, and prove

$$A - \bigcup S = \bigcap T_2$$

$$A - \bigcap S = \bigcup T_2$$

(generalized De Morgan laws).

Proof.

a) First, we introduce a small lemma: For every $X \in S$, $A \cap X \in \mathcal{P}(A)$. Indeed, $A \cap X \subseteq A$, and so by definition of $\mathcal{P}(A)$, $A \cap X \in \mathcal{P}(A)$.

Now we prove that both sets are equal using the Axiom of Extensionality.

First, let $x \in A \cap \bigcup S$. To show that $x \in \bigcup T_1$, we seek an element of T_1 which contains x. since x belongs to $\bigcup S$, $x \in X$ for some $X \in S$. Now consider the set $A \cap X$. By the above lemma, it is an element of $\mathcal{P}(A)$, and thus an element of T_1. Since A and X both contain x, $x \in A \cap X$. Therefore, $A \cap X$ is the set we were seeking.

Conversely, let $x \in \bigcup T_1$. By the Axiom of Union, there exists a $Y \in T_1$ which contains x. Now by the definition of T_1, $Y = A \cap X$ for some $X \in S$. Since $x \in Y = A \cap X$, we have $x \in \bigcup S$, as $x \in X$ for some $X \in S$. Hence $x \in A \cap \bigcup S$, as desired.

Therefore, by the Axiom schema of Extensionality, $A \cap \bigcup S = \bigcup T_1$.

b) 1. $A - \bigcup S = \bigcap T_2$. First, suppose that $x \in A - \bigcup S$. Then we have $x \in A$, while x belongs to none of the sets in S. Now let $Y \in T_2$. By the definition of T_2, there exists an $X \in S$ such that $Y = A - X$. Since $x \in A$ and $x \notin X$, it follows that $x \in Y$. Hence $x \in Y$ for every $Y \in T_2$, and so $x \in \bigcap T_2$.
Conversely, suppose that $x \in \bigcap T_2$. For any $X \in S$, we have $A - X \in T_2$ by definition. Then $x \in A - X$, since x belongs to any element of T_2. Hence for every $X \in S$, $x \in A$ and $x \notin X$. Since there is no $X \in S$ such that $x \in X$, $x \notin \bigcup S$. Therefore $x \in A - \bigcup S$.
By extensionality, $A - \bigcup S = \bigcap T_2$.

2. $A - \bigcap S = \bigcup T_2$.

□

2 Relations, Functions, and Orderings

2.1 Ordered Pairs

2.2 Relations

2.3 Functions

2.4 Equivalences and Partitions

2.5 Orderings

Exercise 2.5.1.

a) Let R be an ordering of A, S be the corresponding strict ordering of A, and R^* be the ordering corresponding to S. Show that $R^* = R$.

b) Let S be a strict ordering of A, R be the corresponding ordering, and S^* be the strict ordering corresponding to R. Then $S^* = S$.

Exercise 2.5.2. Let $\mathrm{Fn}(X, Y)$ be the set of all functions mapping a subset of X into Y. Define a function $\leq$ in $\mathrm{Fn}(X, Y)$ by

$$f \leq g \text{ if and only if } f \subseteq g.$$

a) Prove that $\leq$ is an ordering of $\mathrm{Fn}(X, Y)$.

b) Let $F \subseteq \mathrm{Fn}(X, Y)$. Show that $\sup F$ exists if and only if F is a compatible system of functions; then $\sup F = \bigcup F$.

Proof.

a) The proof is obvious from the properties of $\subseteq$.

b)
1. Suppose that $\sup(F)$ exists, and let $f, g \in F$. Let $x \in \mathrm{dom}(f) \cap \mathrm{dom}(g)$. Since $f \leq \sup(F)$, i.e., $f \subseteq \sup(F)$, we have $f(x) = (\sup(F))(x)$. Similarly, $g(x) = (\sup(F))(x)$, and we conclude that $f(x) = g(x)$.
2. Suppose that F is a compatible system of functions. We claim that $\bigcup F = \sup(F)$.
   i) By Theorem 3.12, $\bigcup F$ is a function. If $x \in \mathrm{dom}(\bigcup F)$, $(x, y) \in f$ for some y and $f \in F$. But then $x \in X$, since $\mathrm{dom}(f) \subseteq X$. So $\mathrm{dom}(\bigcup F) \subseteq X$, and similarly, $\mathrm{ran}(\bigcup F) \subseteq Y$. Hence $\bigcup F \in \mathrm{Fn(X, Y)}$.

ii) Let $f \in F$ and $(x, y) \in f$. Then immediately $(x, y) \in \bigcup F$, so $f \leq \bigcup F$ for every $f \in F$; $\bigcup F$ is an upper bound of F.

iii) Finally, let g be an upper bound of F. If $(x, y) \in \bigcup F$, $(x, y) \in f$ for some $f \in F$. But then $(x, y) \in g$, since $f \subseteq g$. Therefore $\bigcup F \subseteq g$, that is, $\bigcup F \leq g$.

□